

```
1. public class Circle  
{  
    private double radius;  
    private String color;  
    public Circle ()  
}
```



```
this.radius = 1.0;  
this.color = "red";
```

```
}  
public Circle (double radius)
```

```
{  
    this.radius = radius;  
    this.color = "red";
```

```
}  
public Circle (double radius, String color) {
```

```
    this.radius = radius;  
    this.color = color;
```

```
}  
public double getRadius() {  
    return radius;
```

```
}  
public void setRadius (double radius) {  
    this.radius = radius;
```

```
}  
public String getColor() {  
    return color;
```

```
}  
public void setColor (String color) {  
    this.color = color;
```



```

public double getArea() {
    return Math.PI * this.radius * this.radius;
}

public String toString() {
    return "Circle [radius = " + radius + ", color = "
    + color + " ]";
}
}

```

3 Bài 2: Triển khai class Rectangle theo diagram

```

import java.util.Scanner;

public class Rectangle {
    private int length;
    private int width;

    public Rectangle() {
        this.length = 0;
        this.width = 0;
    }

    public Rectangle (int length, int width) {
        this.length = length;
        this.width = width;
    }

    public int getLength() {

```



```
return length;
```

```
public void setLength (int length) {  
    this.length = length;  
}
```

```
public int getWidth () {  
    return width;  
}
```

```
Diagram public void setWidth (int width) {  
    this.width = width;  
}
```

```
public int getArea () {  
    return length * width;  
}
```

@ Override

```
public String toString () {  
    return "Rectangle [length = " + length + "  
    width = " + width + " ]";  
}
```

```
public class Main {  
    public static void main (String [] args) {  
        System.out.println ("Kona Circle");  
    }  
}
```



```
Rectangle rect 1 = new Rectangle (10, 5);  
System.out.println (rect 1.toString());  
System.out.println ("Area " + rect 1.get
```

```
Area());
```

```
rect 1.setLength (20);  
System.out.println (rect 1.toString());  
System.out.println ("New length: " + rect 1
```

```
.getLength  
());
```



```
Rectangle rect 1 = new Rectangle (10, 5);  
System.out.println (rect 1.toString ());  
System.out.println ("Area" + rect 1.get
```

```
rect 1.setlength (20);  
System.out.println (rect 1.toString ());  
System.out.println ("New length: " + rect 1
```

Bài 3. Triển khai class Employee theo diagram.

```
import java.util.Scanner;  
public class Employee {  
    private int id;  
    private String firstName;  
    private String lastName;  
    private int salary;  
}
```

```
    public int getInt () {  
        return id;  
    }
```

```
    public String getFirstName () {  
        return firstName;  
    }
```



```

    public String getLastName() {
        return lastName;
    }

    public int getSalary() {
        return salary;
    }

    public void setSalary(int salary) {
        this.salary = salary;
    }

```

```

    public String getFullName() {
        return lastName + " " + firstName;
    }

```

```

    public int getAnnualSalary() {
        return salary * 12;
    }

```

```

    public int upToSalary(int percent) {
        this.salary = salary + (salary * percent) / 100;
        return this.salary;
    }

```

② Override

```

    public String toString() {
        return "Employee [id = " + id + ", name = " +
            getFullName() + ", salary = " + salary + " ]";
    }

```



```

public class TestEmployee {
    public static void main (String [] args) {
        Employee e1 = new Employee (101, "An",
        "Nguyen Van", 1000);
        System.out.println (e1.toString());
        System.out.println ("Họ tên đầy đủ: "
        + e1.getFullName());
        System.out.println ("Lương Hàng Năm: "
        + e1.getAnnualSalary());
        System.out.println ("Lương sau khi tăng
        10% : " + e1.upToSalary (10));
        System.out.println ("Thông tin mới: "
        + e1.toString());
    }
}

```


Bài 4: Triển khai class Account theo

diagram

```
import java.util.Scanner;
```

```
public class Account {
```

```
    private String id;
```

```
    private String name;
```

```
    private int balance;
```



```

public Account (String id, String name, int balance) {
    this.id = id;
    this.name = name;
    this.balance = balance;
}

public String getId() {
    return id;
}

public String getName() {
    return name;
}

public int getBalance() {
    return balance;
}

public int credit (int amount) {
    if (amount > 0) {
        this.balance += amount;
    }
    return this.balance;
}

public int debit (int amount) {
    if (amount <= balance) {
        this.balance = amount;
    } else {

```

diagram


```
System.out.println("Thanh toán không  
thành công!");
```

```
return this.balance;
```

```
public int transferTo (Account another,  
int amount) {
```

```
if (amount <= balance) {
```

```
this.balance -= amount;
```

```
} else {
```

```
System.out.println("Chuyển tiền không  
thành công!");
```

```
return this.balance;
```

© Override

```
public String toString() {
```

```
return "Account [id = " + id + ", name  
= " + name + ", balance = " + balance + "];"
```

```
}
```

```
public class Main {
```

```
public static void main (String [] args) {
```



```
Account a = new Account("A01", "Account A");  
Account b = new Account("A02", "Account B");  
System.out.println("Ban dau:");  
System.out.println(a.toString());  
System.out.println(b.toString());  
System.out.println("Sau khi thay doi:");  
System.out.println(a.toString());  
System.out.println(b.toString());
```